

UNIT 06

STATE OF MATTER (III): SOLIDS



Simple Properties of Solids

Before going to deal with the concept of solid in broader aspect in the next sections, you need to understand some simple properties of solids related to their inter-particles' associations based on some limitations in the freedom of particles movement.



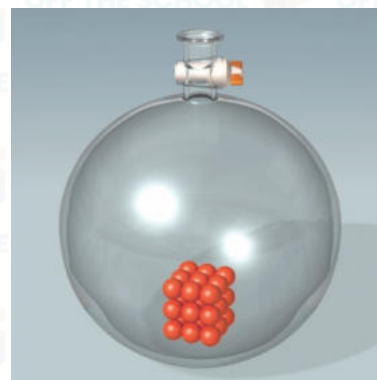
Gas

Particles in a gas are separated by large distances and free to move entirely independently of one another.



Liquid

Particles in a liquid are touching one another but free to move about.



Solid

Particles in a solid are essentially locked in place with respect to one another.

Kinetic energy and Molecular motion:

Particles of solid can only vibrate about their mean position. They do not bear translational and rotational kinetic energy as liquid particles do.

Intermolecular forces:

Particles in solid held together tightly in a fixed position due to some strong intermolecular forces.

Compression:

Solids are nearly incompressible. In term of kinetic theory, particles are tightly bonded together due to strong attractive forces therefore only a very small empty space is left among them. It makes the solid denser, rigid and hard and can withstand considerable external pressure.

Diffusion:

Solids diffuses very slowly as compared to liquids and gases. This is due to the close packing of particles and restriction against their free movement.

Expansion:

Solids expand on heating but this ability is very little as compared to liquids and gases. Since particles of solids are arranged orderly in a fixed position, an increase in temperature weakens their attractive forces giving an increase in the volume.

Melting:

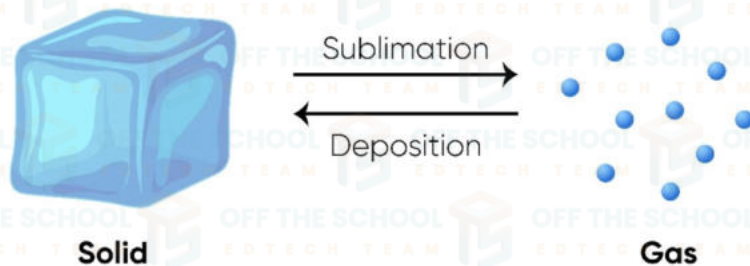
Solids melt on heating at a particular temperature. In term of kinetic theory, vibrational energy of particles increases on heating until at melting temperature the vibrational energy of the solid particles overcome the energy of existing bonds among them thus solid melt and change into liquid state.

Sublimation:

Some solids directly change into vapors on heating without passing through liquid phase, which is known as sublimation.

For example: naphthalene, iodine, camphor and solid CO_2 are sublime. In term of kinetic theory, the attractive forces in these solids are quite weaker compared with ordinary solids, thus on heating high energy surface particles of these solids directly pass into vapors.



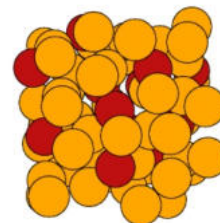


TYPES OF SOLIDS

Most of the substances around us are solids rather than liquid or gases. Solids like sugar, salts, metals etc. are hard and crystalline while some others such as rubber, plastic, etc. are soft and amorphous. There are two main classes of solids named as crystalline solids and amorphous solids.

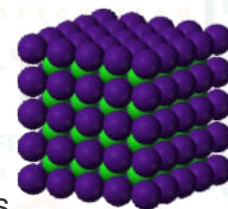
Amorphous Solids:

These are solids without a regular particle arrangement, leading to a lack of defined shape and a broad melting range rather than a specific melting point. Examples of amorphous solids are rubber, glass, and plastics. They gradually soften and behave like supercooled liquids upon heating.



Crystalline Solids:

They are substances with particles arranged in a well-defined, three-dimensional pattern, known as a crystal lattice. This structure gives crystalline solids a specific geometric shape and a precise melting point. Common examples include table salt and diamonds.



Properties of Crystalline Solids

1. Symmetry: Crystalline solids display symmetry, defined by regular repetitive patterns of edges and faces when rotated. This symmetry consists of planes, axis, and centers of symmetry.

- **Plane of symmetry:** An imaginary plane that bisects the crystal into two equal parts, each a mirror image of the other.
- **Axis of symmetry:** An imaginary line around which the crystal can be rotated 360 degrees to appear undisturbed.
- **Centre of symmetry:** A point within the crystal from which any straight line will intersect equal surfaces on opposite sides.

2. Geometrical Shape: Crystalline solids have an orderly and repeating geometric shape due to their organized particle arrangement in three dimensions, such as the hexagonal shape of a diamond's unit cell.

3. Sharp Melting Point: They melt at specific temperatures due to uniform long-range attractive forces between particles.

4. Cleavage Plane: They can be split along specific planes, known as cleavage planes, producing smaller crystals of identical shape.

Anisotropy:

Anisotropy means having properties that vary depending on the direction along which they are measured. Crystalline solids often exhibit anisotropy because their atoms are arranged in an orderly pattern.



Isomorphism:

A phenomenon where different solid substances can exist in the same crystalline structure, referred to as isomorphism. The substances, although chemically different, have atoms in the same atomic ratio and display similar shapes but different physical and chemical properties.

Polymorphism:

A characteristic of crystalline solids where one chemical compound can have more than one crystal structure. Polymorphism implies that the same chemical substance can exist in different crystalline forms, which may exhibit varied physical properties like melting point, structure, and hardness.

Allotropic Modification:

Allotropy refers to the existence of an element in two or more different crystalline structures, which have the same chemical nature but different physical properties. A classic example is carbon, which can exist as diamond with a cubic structure and graphite with a hexagonal structure. For sulphur, two important allotropic forms are rhombic sulphur (stable at ordinary temperatures) and monoclinic sulphur (stable above 95.5°C).

**Transition temperature**

Transition temperature is the temperature at which a substance changes from one physical state to another, like from solid to liquid (melting point), or liquid to gas (boiling point). Example; Ice melting at 0°C.

Types of crystalline solids:

Ionic Solids: Crystalline solids made up of positively and negatively charged ions. The ions are arranged in a repeating three-dimensional lattice. This arrangement creates strong and long-range electrostatic attractions in all directions, giving ionic solids their distinct properties.

1
A photograph showing crystals of sodium chloride

2
A model of a portion of a crystal detailing the regular arrangement of sodium ions and chloride ions. Each sodium ion is surrounded by six chloride ions, and each chloride ion is surrounded by six sodium ions.

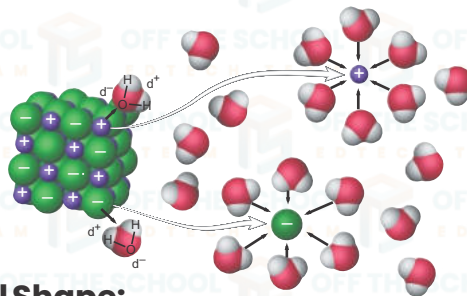
3
A model highlighting the packing arrangement of the sodium and chloride ions in the crystal.

▲ The sodium chloride crystal



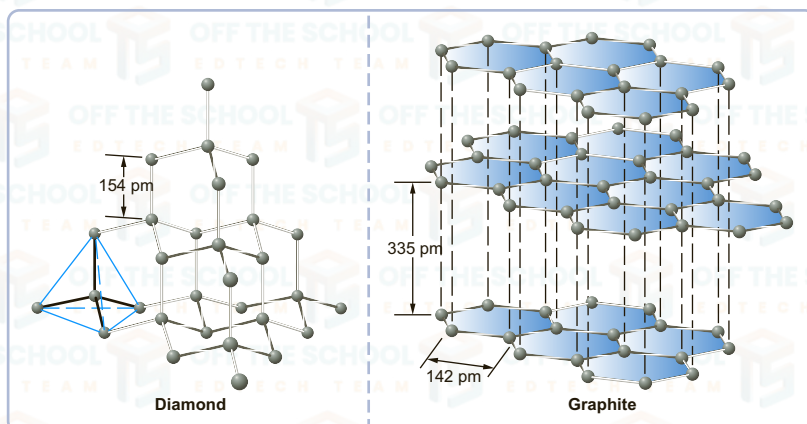
Properties of Ionic Solids:

- 1. Hardness:** Ionic solids are hard at room temperature due to strong electrostatic forces.
- 2. Brittleness:** They are brittle and can shatter when subjected to force.
- 3. High Melting Point:** The strong ionic bonds give them high melting points.
- 4. Solubility:** They are typically soluble in water and other polar solvents.
- 5. Conductivity:** Ionic solids do not conduct electricity in their solid state but can conduct when molten or dissolved in water due to free-moving ions.

**Radius Ratio and Crystal Shape:**

- A small radius ratio leads to a lower coordination number and a simpler geometric shape, while a larger ratio leads to a higher coordination number and a more complex shape.

Covalent Solids: These consist of atoms connected by covalent bonds into a large, single network. Examples include diamond, where each carbon atom is tetrahedrally bonded to four others, and graphite, with layers of hexagonally arranged carbon atoms held together by weak forces. Covalent solids are typically hard, with high melting points due to strong bond networks.

**Structures of diamond and graphite**

Diamond is a three-dimensional network solid: every carbon atom is covalently bonded to four others. Graphite consists of sheets of carbon atoms covalently bonded to form a hexagonal pattern of atoms; the sheets are held together by van der Waals forces.

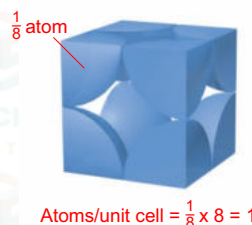
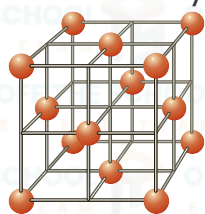
Metallic Solids: Composed of metal ions surrounded by a "sea" of delocalized electrons. This structure gives metallic solids their characteristic properties, such as electrical conductivity, malleability, ductility, and a shiny appearance known as metallic luster.

Molecular Solids: Made up of molecules bonded by weaker intermolecular forces like hydrogen bonds, dipole-dipole interactions, or London dispersion forces. These solids tend to have low melting points and are usually poor conductors of electricity.

The properties of these solids are dictated by the type of particles they contain and the nature of the bonds between these particles, leading to diverse physical characteristics.

Crystal Lattice: A regular, repeating arrangement of atoms, ions, or molecules in three-dimensional space, forming a structure called a crystal lattice or space lattice. It's made up of infinite points representing the positions of particles within the crystal.

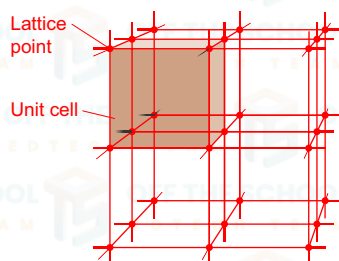
Unit Cell: The simplest repeating unit in a crystal lattice that outlines the entire structure's pattern. It's the smallest portion from which the lattice is built and is repeated to form the entire crystal.



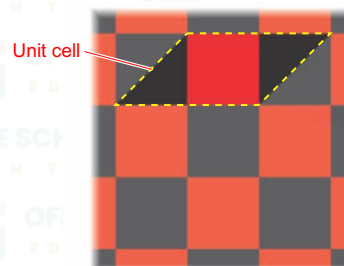
$$\text{Atoms/unit cell} = \frac{1}{8} \times 8 = 1$$

Unit Cell of NaCl Crystal: NaCl forms a cubic crystal lattice. The Na^+ and Cl^- ions are arranged in such a way that each sodium ion is surrounded by six chloride ions, suggesting a coordination number of 6 for Na^+ in the lattice.

The crystal lattice and the unit cell. **A**, A small portion of a lattice is shown as points connected by lines, with a unit cell (colored). **B**, A checkerboard analogy for a lattice.



A Portion of 3-D lattice



B 2-D analogy for unit cell and lattice

Lattice Energy: The energy released when oppositely charged ions in a gaseous state form a solid ionic lattice. It can also be viewed as the energy required to separate a mole of a solid ionic compound into its gaseous ions.

Lattice energy is affected by the size and charge of the ions involved, with smaller ions and ions of higher charge typically having higher lattice energy.

Difference between crystalline and amorphous solids:

Crystalline Solids	Amorphous Solids
Have a highly ordered structure, where atoms are arranged in a repeating pattern extending in all three dimensions.	Lack a long-range ordered structure; atoms or molecules are arranged more randomly than in crystalline materials.
Exhibit a sharp melting point; they transition from solid to liquid at a specific temperature.	Do not have a precise melting point; they soften over a range of temperatures.
Are often anisotropic, meaning their properties differ based on the direction along the crystal lattice.	Are usually isotropic, which means their properties are the same in all directions.
Show well-defined edges and faces when cleaved or fractured.	When broken, they fracture with a curvy, irregular surface rather than a flat plane.
Examples include table salt, diamonds, and quartz.	Examples include glass, plastics, and gels.

Difference between isomorphism and polymorphism:

Isomorphism	Polymorphism
It refers to the phenomenon where two or more distinct compounds share the same crystal structure, or lattice type, due to having a similar arrangement of atoms.	Polymorphism is the ability of a single chemical substance to exist in two or more different crystalline forms, known as polymorphs.

Isomorphous substances have different chemical compositions but form crystals with similar geometric shapes because of their similar ionic sizes and charges.	Polymorphism refers to the ability of a chemical substance to exist in more than one crystalline form, called polymorphs.
They can replace each other within the same crystal lattice without altering the crystal structure.	Each polymorph of a substance has distinct physical properties such as melting point, hardness, and structure, despite having the same chemical composition.
An example is the ability of KCl (potassium chloride) and NaCl (sodium chloride) to form isomorphous crystals	A common example is carbon, which exists as graphite and diamond; both are pure carbon but have different physical properties due to their different crystal structures.

Scientific Reason(s):**1. Explain why diamond is a non-conductor of electricity and graphite is a conductor.**

Diamond doesn't conduct electricity because its electrons are tightly bound in a lattice. In contrast, graphite conducts electricity because it has free electrons in its layered structure that can move and carry a charge.

2. Why the compressibility of Solids is nearly zero?

Solids are hard to compress because their particles are tightly packed together.

3. Why some solids are sublime in nature?

Some solids can directly change into a gas because they have strong surface forces but weak forces holding the bulk material together.

4. Give a brief account on metallic solids. Why they conduct heat and electricity?

Metallic solids are made of metal atoms packed closely together. They conduct heat and electricity well because their electrons can move freely throughout the metal lattice. This "sea of electrons" allows energy to pass quickly from one part to another.

5. What is meant by polymorphism? How is it related to Allotropy? Give your Answer with examples.

Polymorphism refers to the ability of a solid material to exist in more than one form or crystal structure. Allotropy is a type of polymorphism that occurs in chemical elements. For instance, carbon is polymorphic because it can exist as diamond or graphite, which are allotropes of carbon.

6. Molecular solids are soft and possess low melting point, how can you justify this statement?

Molecular solids are generally soft and have low melting points because the molecules are held together by weak intermolecular forces, such as van der Waals forces or hydrogen bonds. These forces require less energy to overcome, leading to a lower melting point.

